

This project is a turn-based battle damage optimizer, taking inspiration from attacks in One Piece. The program determines the most devastating combination that can be made, given restricted mana, stamina, and turns. We read attack data from a CSV file and store it in C++ data structures, including vectors and structs. Then it goes back and forth and uses these strategies and tries other attack combinations. The software runs and finds the best possible series of hits and writes out the best combo and the most damage it can deal. In CS 311 we learned about filing, recursion, vectors, optimization logic, algorithms, etc. This project shows how data structures and algorithms can be utilized to tackle optimization problems in the style of video games.